

Stage 3 Pilot Results — Sponsorship Network

February 13, 2026

Overview

Dependent Variables

Primary DV: Trust Change (Decision 2 - Decision 1)

- Used in RQ1 and RQ4
- Measures how evaluators update their trust after observing an outcome

Secondary DVs:

- **Absolute Endorsement Strength** (RQ2): $|\text{Rating} - 50|$, measures sponsor endorsement behavior
- **Initial Trust** (RQ3): Decision 1 trust, measures evaluator's baseline response to endorsements

Study Design

Each evaluator sees ONE sponsor making TWO decisions:

1. **Decision 1:** Evaluator sees sponsor endorsement (with strength) $\rightarrow$ stakes $X\%$ trust $\rightarrow$ learns outcome (success/failure)
2. **Decision 2:** SAME evaluator sees SAME sponsor on new candidate $\rightarrow$ stakes $Y\%$ trust
3. **Trust Change** = $Y\% - X\%$

Research Questions

- **RQ1/HQ1:** Are female sponsors punished **more** for failures and rewarded **less** for successes?
- **RQ2/HQ2:** Do female sponsors provide **weaker** endorsements than male sponsors?
- **RQ3/HQ3:** Do female sponsors need **higher** endorsement strength to achieve same initial trust?
- **RQ4/HQ4:** Is trust change **more** contingent on endorsement strength $\times$ outcome for female sponsors?

Pilot Sample

Filter Step	N Remaining
Total export rows	277
Exclude preview/incomplete	271
Valid PROLIFIC_PID	269
Target study (698e27db580428e78f8e9ff1)	234
Pass comprehension (Q42=2)	100

Filter Step	N Remaining
Has condition + Q1 stake	100

Final analytic sample: N = 100

Endorser gender: Man = 48 | Woman = 52

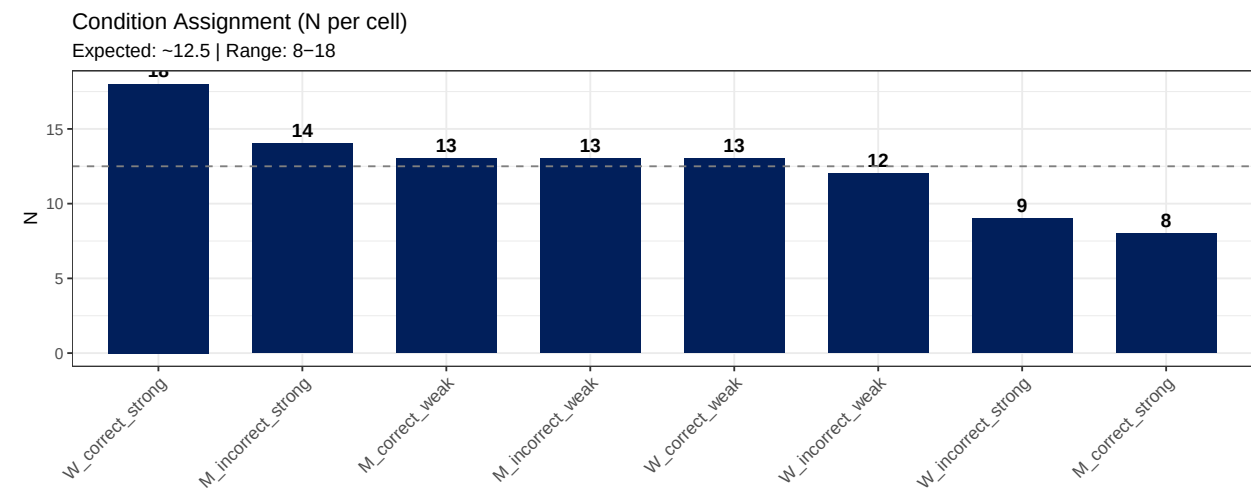
Correctness: Correct = 52 | Incorrect = 48

Strength: Strong = 49 | Weak = 51

Trust change (Q2): N = 100

Median completion: 5.6 min

Gender manipulation check accuracy: 82.6 %



RQ1: Trust Change by Sponsor Gender $\times$ Protege Outcome

Hypothesis (HQ1): Female sponsors are punished **more** for failures and rewarded **less** for successes.

Model

$$\text{trust_change}_i = \beta_0 + \beta_1 \text{Woman}_i + \beta_2 \text{Success}_i + \beta_3 (\text{Woman}_i \times \text{Success}_i) + \beta_4 \text{abs_strength_q1_c}_i + \epsilon_i$$

Variable definitions:

Term in equation	R output name	Levels (reference in parentheses)
Woman_i	endorser_gender_fWoman	(Man = 0), Woman = 1
Success_i	outcome_fSuccess	(Failure = 0), Success = 1
$\text{Woman} \times \text{Success}$	endorser_gender_fWoman:outcome_fSuccess	Interaction
$\text{abs_strength_q1_c}_i$	abs_strength_q1_c	Mean-centered

- β_3 tests HQ1: negative β_3 means the success boost is **smaller** for female sponsors

```
##
## Call:
## lm(formula = trust_change ~ endorser_gender_f * outcome_f + abs_strength_q1_c,
##     data = df_tc)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -115.269  -20.197   -1.215   23.744   78.210
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      -5.7042     7.4785  -0.763  0.44750
## endorser_gender_fWoman      3.8979    11.3250   0.344  0.73147
## outcome_fSuccess     21.6298    11.3558   1.905  0.05984 .
## abs_strength_q1_c      -0.5026     0.1613  -3.116  0.00243 **
## endorser_gender_fWoman:outcome_fSuccess -16.6779    15.9289  -1.047  0.29775
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 38.84 on 95 degrees of freedom
## Multiple R-squared:  0.1359, Adjusted R-squared:  0.09948
## F-statistic: 3.734 on 4 and 95 DF,  p-value: 0.007246
```

Interpretation

- β_0 (**Intercept**): Mean trust change for **male sponsors after failure**, at average endorsement strength
- β_1 (**endorser_gender_fWoman**): How much trust change differs for **female sponsors after failure** (relative to male)
- β_2 (**outcome_fSuccess**): The success boost for **male sponsors** — how much more trust they get after success vs. failure
- β_3 (**endorser_gender_fWoman:outcome_fSuccess**): **HQ1 test** — how much **smaller** (if negative) the success boost is for female sponsors
- β_4 (**abs_strength_q1_c**): Controls for the pretreatment strength of the endorsement

Visualization

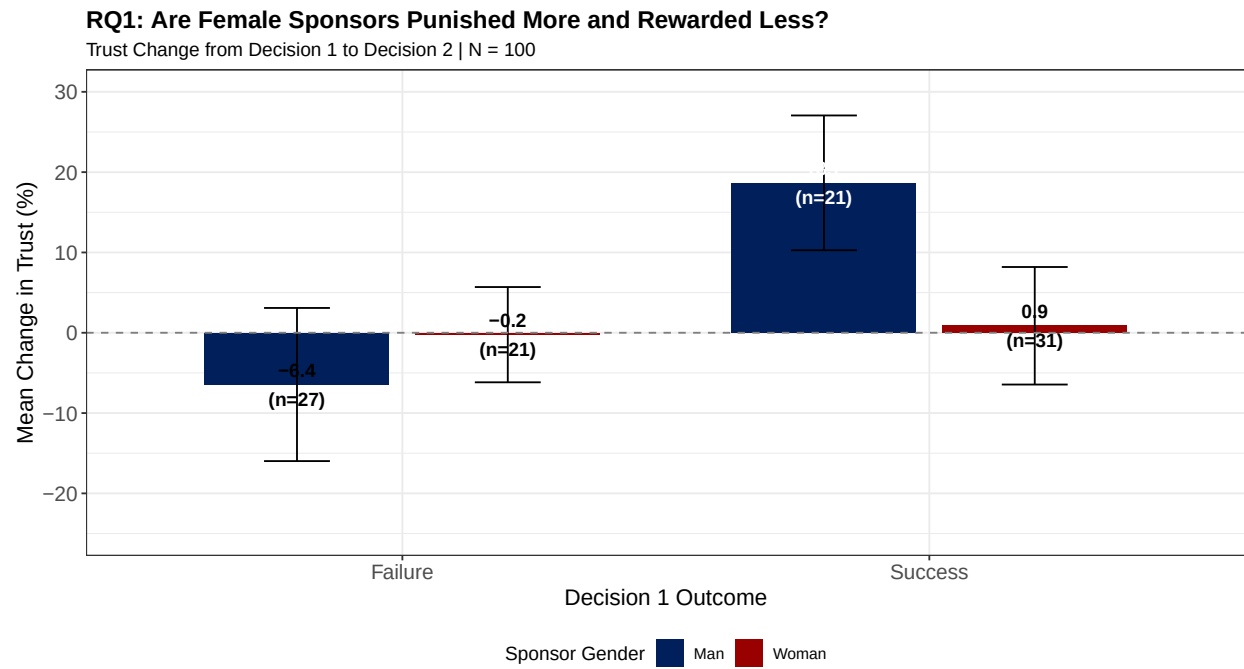


Figure 1: RQ1: Trust Change by Sponsor Gender and Decision 1 Outcome

RQ2: Endorsement Strength by Sponsor Gender $\times$ Protege Gender

Hypothesis (HQ2): Female sponsors provide **weaker** endorsements than male sponsors.

Model

$$\text{abs_strength}_{ij} = \beta_0 + \beta_1 \text{Woman_Sponsor}_i + \beta_2 \text{Woman_Protege}_j + \beta_3 (\text{Woman_Sponsor}_i \times \text{Woman_Protege}_j) + u_i + \epsilon_{ij}$$

Variable definitions:

Term in equation	R output name	Levels (reference in parentheses)
Woman_Sponsor _i	endorser_gender_fWoman	(Man = 0), Woman = 1
Woman_Protege _j	endorsed_gender_fWoman	(Man = 0), Woman = 1
u _i	(1 PROLIFIC_PID)	Random intercept per endorser

- β_1 tests HQ2: negative β_1 means female sponsors give weaker (closer to neutral) endorsements
- Mixed model with random intercepts accounts for 10 repeated trials per endorser

This analysis uses Stage 2 main study data (N = 196 endorsers, 1960 trial observations).

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## abs_strength ~ endorser_gender_f * endorsed_gender_f + (1 | PROLIFIC_PID)
## Data: s2_model_data
##
## REML criterion at convergence: 15516
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.5913 -0.8472  0.1647  0.8538  1.6969
##
## Random effects:
## Groups      Name                Variance Std.Dev.
## PROLIFIC_PID (Intercept)    36.72      6.06
## Residual                    228.16    15.11
## Number of obs: 1856, groups: PROLIFIC_PID, 196
##
## Fixed effects:
##
##              Estimate Std. Error      df
## (Intercept)      31.0734      0.9327  391.2249
## endorser_gender_fWoman      1.3707      1.3359  388.6485
## endorsed_gender_fWoman     -0.2927      0.9932 1716.1182
## endorser_gender_fWoman:endorsed_gender_fWoman      0.1356      1.4208 1710.2377
##
##              t value Pr(>|t|)
## (Intercept)      33.317  <2e-16 ***
## endorser_gender_fWoman      1.026   0.306
## endorsed_gender_fWoman     -0.295   0.768
## endorser_gender_fWoman:endorsed_gender_fWoman      0.095   0.924
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Correlation of Fixed Effects:
##      (Intr) endrsr__W endrsd__W
## endrsr_gn_W -0.698
## endrsd_gn_W -0.552  0.385
## endr__W:__W  0.386 -0.550   -0.699
```

Visualization

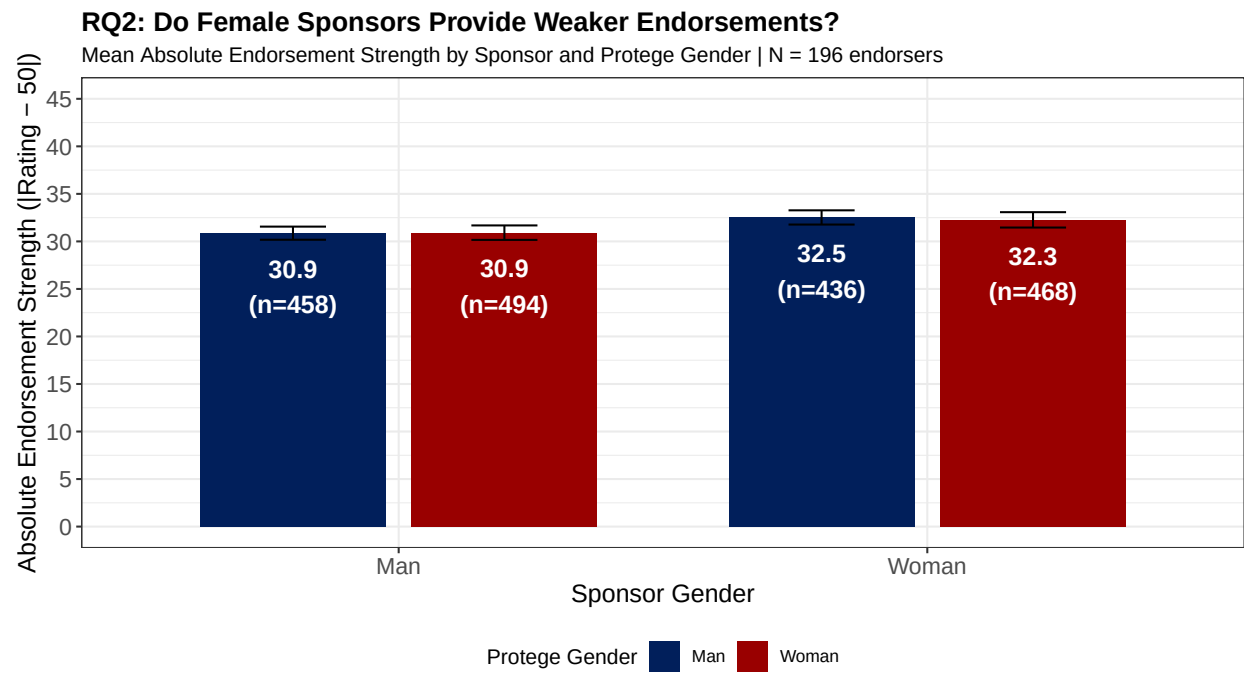


Figure 2: RQ2: Absolute Endorsement Strength by Sponsor and Protege Gender

RQ3: Initial Trust by Sponsor Gender × Endorsement Strength

Hypothesis (HQ3): Female sponsors need **higher** endorsement strength to achieve same initial trust as males.

Model

$$\text{stake_percent_q1}_i = \beta_0 + \beta_1 \text{Woman}_i + \beta_2 \text{abs_strength_q1}_i + \beta_3 (\text{Woman}_i \times \text{abs_strength_q1}_i) + \epsilon_i$$

Variable definitions:

Term in equation	R output name	Levels / range
stake_percent_q1 _i	stake_percent_q1	0–100 (evaluator’s initial trust)
Woman _i	endorser_gender_fWoman	(Man = 0), Woman = 1
abs_strength_q1 _i	abs_strength_q1	

- β_2 : How much initial trust increases per unit of endorsement strength (for **male** sponsors)
- β_3 tests HQ3: negative β_3 means female sponsors have a **shallower** slope (need more strength for same trust)

```
##
## Call:
## lm(formula = stake_percent_q1 ~ endorser_gender_f * abs_strength_q1,
##     data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -62.053 -15.066   2.704  12.947  53.936
##
## Coefficients:
##                                Estimate Std. Error t value Pr(>|t|)
## (Intercept)                   46.9769     4.7452   9.900 2.44e-16 ***
## endorser_gender_fWoman          -1.2392     6.7806  -0.183 0.855380
## abs_strength_q1                 0.5615     0.1401   4.007 0.000121 ***
## endorser_gender_fWoman:abs_strength_q1 -0.2352     0.1942  -1.212 0.228658
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 23.61 on 96 degrees of freedom
## Multiple R-squared:  0.1964, Adjusted R-squared:  0.1713
## F-statistic: 7.821 on 3 and 96 DF,  p-value: 0.0001005
```

Interpretation

- β_0 (**Intercept**): Predicted initial trust for **male sponsors** when abs_strength = 0
- β_1 (**endorser_gender_fWoman**): Baseline trust difference for female sponsors at zero strength
- β_2 (**abs_strength_q1**): Slope for **male** sponsors — trust gained per unit of strength
- β_3 (**endorser_gender_fWoman:abs_strength_q1**): **HQ3 test** — negative means females have a shallower slope

Visualization

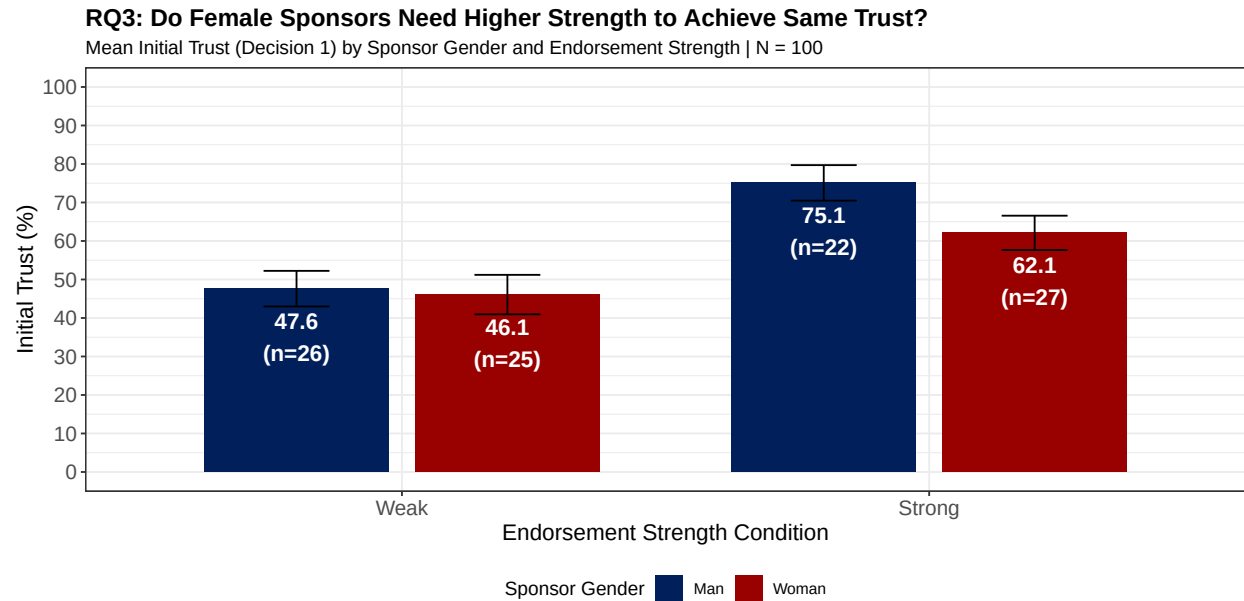


Figure 3: RQ3: Initial Trust by Sponsor Gender and Endorsement Strength

Observed pattern:

- At **Weak** endorsement strength, both genders start at similar initial trust (~48% for males, ~46% for females)
- At **Strong** endorsement strength, males receive **75.1%** trust vs. females at **62.1%** — a 13 pp gap
- The strong endorsement “boost” is **27.5 pp** for males but only **16.0 pp** for females
- The interaction is not significant ($p = .229$), but the direction is consistent with HQ3: female sponsors get less trust per unit of endorsement strength

RQ4: Trust Change by Gender $\times$ Strength $\times$ Outcome (3-Way Interaction)

Hypothesis (HQ4): Trust change is **more** contingent on endorsement strength $\times$ outcome for female sponsors.

Model

$$\begin{aligned} \text{trust_change}_i = & \beta_0 + \beta_1 \text{Woman}_i + \beta_2 \text{abs_str_c}_i + \beta_3 \text{Success}_i \\ & + \beta_4 (\text{Woman} \times \text{abs_str_c})_i \\ & + \beta_5 (\text{Woman} \times \text{Success})_i \\ & + \beta_6 (\text{abs_str_c} \times \text{Success})_i \\ & + \beta_7 (\text{Woman} \times \text{abs_str_c} \times \text{Success})_i + \epsilon_i \end{aligned}$$

Variable definitions:

Term in equation	R output name	Levels / range
Woman	endorser_gender_fWoman	(Man = 0), Woman = 1
abs_str_c	abs_strength_q1_c	Mean-centered
Success	outcome_fSuccess	(Failure = 0), Success = 1
β_7 (3-way)	endorser_gender_fWoman:abs_strength_q1_c:outcome_fSuccess	

- β_7 tests HQ4: positive β_7 means the strength $\times$ outcome interaction is **stronger** for female sponsors

```
##
## Call:
## lm(formula = trust_change ~ endorser_gender_f * abs_strength_q1_c *
##     outcome_f, data = df_tc)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -116.197  -19.981   -0.752   23.373   78.477
##
## Coefficients:
##                                     Estimate Std. Error
## (Intercept)                        -5.68748     7.60377
## endorser_gender_fWoman                4.23782    11.53724
## abs_strength_q1_c                   -0.51403     0.31242
## outcome_fSuccess                    21.31287    11.65212
## endorser_gender_fWoman:abs_strength_q1_c    0.12571     0.47283
## endorser_gender_fWoman:outcome_fSuccess  -16.57120    16.21807
## abs_strength_q1_c:outcome_fSuccess    -0.04366     0.47793
## endorser_gender_fWoman:abs_strength_q1_c:outcome_fSuccess -0.10304     0.66469
##
##                                     t value Pr(>|t|)
## (Intercept)                        -0.748    0.4564
## endorser_gender_fWoman                0.367    0.7142
## abs_strength_q1_c                   -1.645    0.1033
## outcome_fSuccess                     1.829    0.0706
## endorser_gender_fWoman:abs_strength_q1_c    0.266    0.7909
## endorser_gender_fWoman:outcome_fSuccess  -1.022    0.3096
```

```
## abs_strength_q1_c:outcome_fSuccess          -0.091    0.9274
## endorser_gender_fWoman:abs_strength_q1_c:outcome_fSuccess -0.155    0.8771
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 39.44 on 92 degrees of freedom
## Multiple R-squared:  0.1372, Adjusted R-squared:  0.07153
## F-statistic:  2.09 on 7 and 92 DF,  p-value: 0.05234
```

Interpretation

- β_3 (**outcome_fSuccess**): The success boost for **male** sponsors at average endorsement strength
- β_5 (**endorser_gender_fWoman:outcome_fSuccess**): How much smaller the success boost is for **female** sponsors (same as RQ1 but without the abs_strength control)
- β_6 (**abs_strength_q1_c:outcome_fSuccess**): Whether stronger endorsements amplify the success boost for **males**
- β_7 (**3-way**): **HQ4 test** — whether that amplification is different for **females**

Visualization

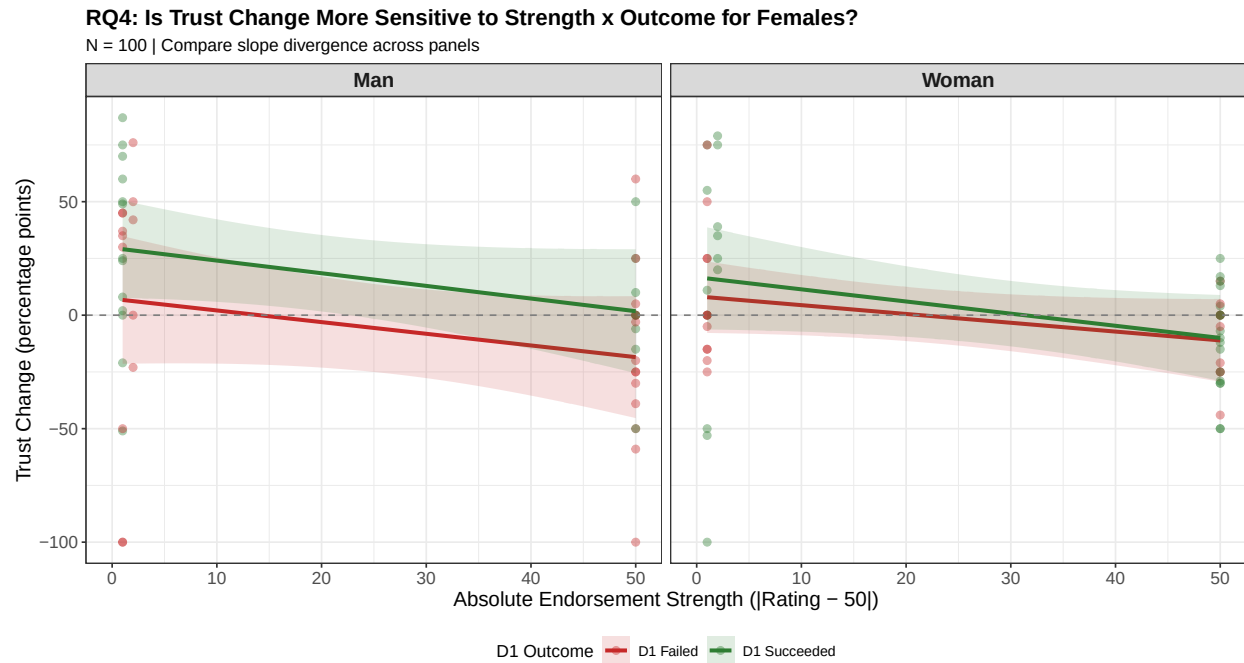


Figure 4: RQ4: Trust Change by Sponsor Gender, Absolute Endorsement Strength, and Outcome

Observed pattern:

- The 3-way interaction is not significant ($p = .877$), indicating that the strength $\times$ outcome relationship does **not** meaningfully differ by sponsor gender
- **All four slope estimates are negative** (ranging from -0.39 to -0.56), meaning that across genders and outcomes, higher endorsement strength is associated with *more negative* trust change
- **Male sponsors** show a large outcome effect (21 pp Success vs. Failure boost, $p = .109$) but no strength $\times$ outcome interaction within males ($p = .935$)
- **Female sponsors** show essentially no outcome effect (5 pp boost, $p = .635$) and no strength $\times$ outcome interaction within females ($p = .719$)
- The core gender asymmetry from RQ1 persists here — male sponsors' outcomes matter, female sponsors' outcomes do not — but this is not moderated by endorsement strength

Appendix: Robustness — Excluding Manipulation Check Failures

Participants were asked to identify the sponsor's gender after the task (Q44). Overall accuracy was 82.6%. Below we re-run the main analyses excluding participants who failed this check or did not respond.

```
## Manipulation check pass: N = 76 of 100
```

```
##   Man: 38 | Woman: 38
```

```
##   Excluded: 24 (Man: 10 | Woman: 14 )
```

Appendix RQ1: Trust Change (Robustness)

```
##
## Call:
## lm(formula = trust_change ~ endorser_gender_f * outcome_f + abs_strength_q1_c,
##     data = df_pass_tc)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -111.266  -20.117   -0.532   26.926   88.552
##
## Coefficients:
##                                Estimate Std. Error t value Pr(>|t|)
## (Intercept)                   -17.4724     9.1429  -1.911   0.0600 .
## endorser_gender_fWoman           12.2529    14.2517   0.860   0.3928
## outcome_fSuccess                29.3501    13.3331   2.201   0.0310 *
## abs_strength_q1_c              -0.4316     0.1935  -2.230   0.0289 *
## endorser_gender_fWoman:outcome_fSuccess -22.9340    19.1946  -1.195   0.2361
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 40.76 on 71 degrees of freedom
## Multiple R-squared:  0.1373, Adjusted R-squared:  0.08874
## F-statistic: 2.826 on 4 and 71 DF,  p-value: 0.031
```

Appendix RQ1: Excluding Manipulation Check Failures

Trust Change | N = 76 (manipulation check passers only)

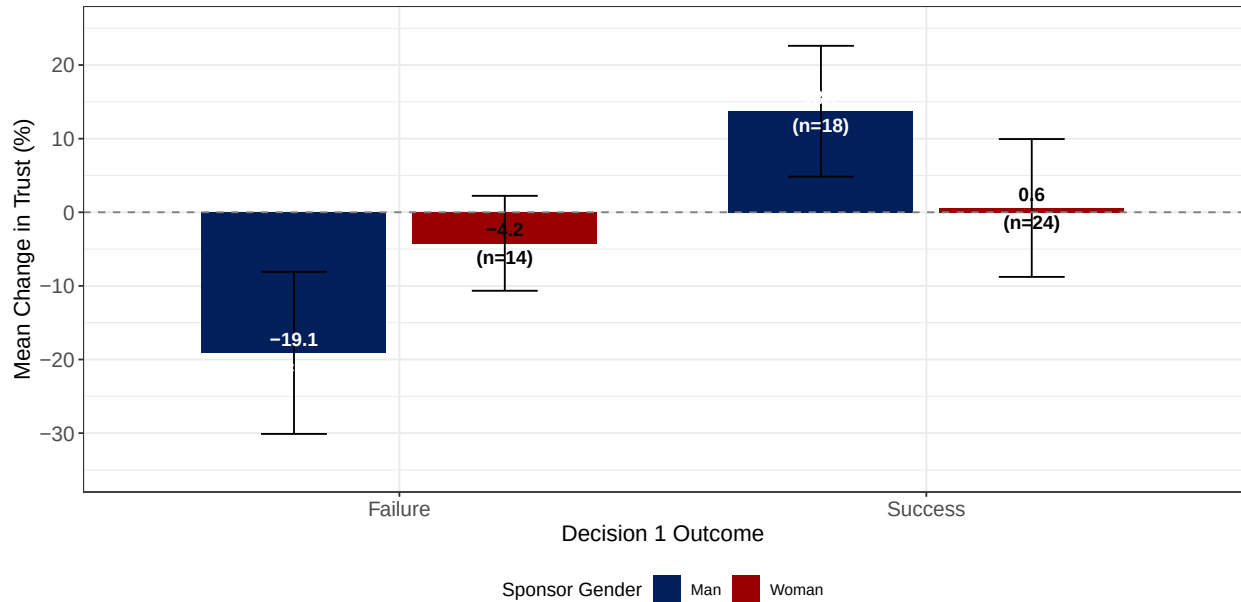


Figure 5: Appendix RQ1: Trust Change — Manipulation Check Passers Only

Appendix RQ3: Initial Trust (Robustness)

```
##
## Call:
## lm(formula = stake_percent_q1 ~ endorser_gender_f * abs_strength_q1,
##     data = df_pass)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -60.284 -18.074   0.849  14.716  51.125
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      49.6573     5.4644   9.087 1.44e-13 ***
## endorser_gender_fWoman      -1.0154     7.7400  -0.131  0.89599
## abs_strength_q1       0.4941     0.1587   3.112  0.00266 **
## endorser_gender_fWoman:abs_strength_q1    -0.2612     0.2248  -1.162  0.24913
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 23.87 on 72 degrees of freedom
## Multiple R-squared:  0.1591, Adjusted R-squared:  0.1241
## F-statistic: 4.541 on 3 and 72 DF,  p-value: 0.005687
```

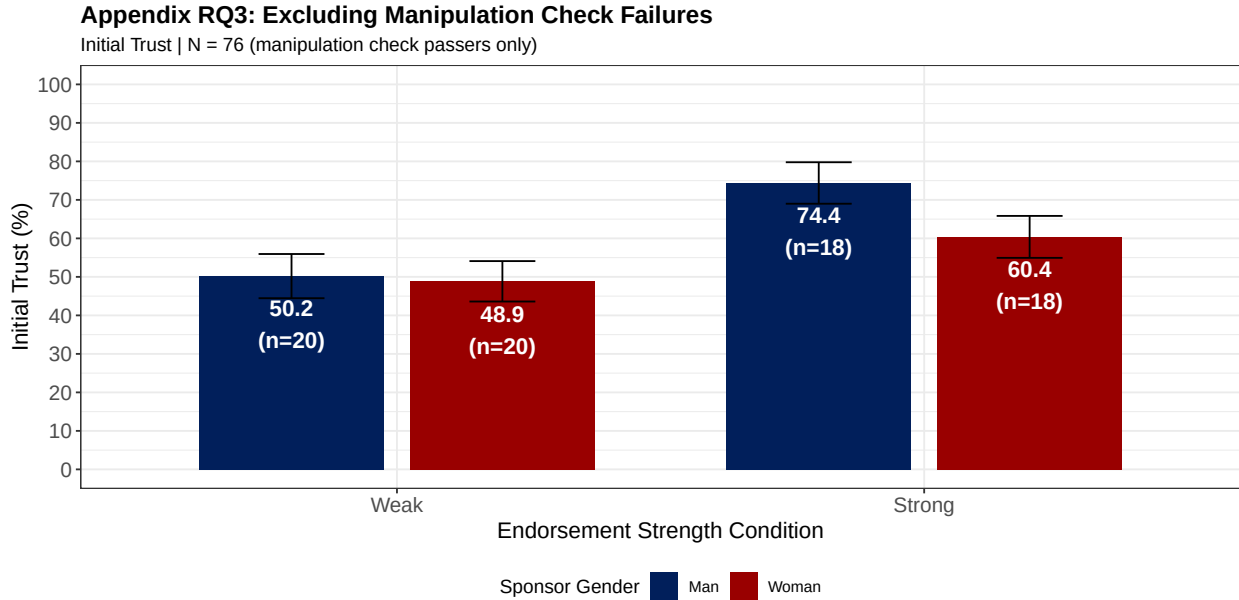


Figure 6: Appendix RQ3: Initial Trust — Manipulation Check Passers Only

Appendix RQ4: Trust Change 3-Way (Robustness)

```
##
## Call:
## lm(formula = trust_change ~ endorser_gender_f * abs_strength_q1_c *
##     outcome_f, data = df_pass_tc)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -117.636  -17.679   -0.571   25.049   83.799
##
## Coefficients:
##                                     Estimate Std. Error
## (Intercept)                       -18.29091     9.35853
## endorser_gender_fWoman              13.40131    14.52113
## abs_strength_q1_c                   -0.21455     0.38190
## outcome_fSuccess                   30.13637    13.62468
## endorser_gender_fWoman:abs_strength_q1_c -0.07541     0.59464
## endorser_gender_fWoman:outcome_fSuccess -23.68431    19.50421
## abs_strength_q1_c:outcome_fSuccess    -0.22462     0.55888
## endorser_gender_fWoman:abs_strength_q1_c:outcome_fSuccess -0.17443     0.80078
##
##                                     t value Pr(>|t|)
## (Intercept)                       -1.954    0.0548 .
## endorser_gender_fWoman              0.923    0.3593
## abs_strength_q1_c                   -0.562    0.5761
## outcome_fSuccess                    2.212    0.0303 *
## endorser_gender_fWoman:abs_strength_q1_c -0.127    0.8995
## endorser_gender_fWoman:outcome_fSuccess -1.214    0.2288
## abs_strength_q1_c:outcome_fSuccess    -0.402    0.6890
## endorser_gender_fWoman:abs_strength_q1_c:outcome_fSuccess -0.218    0.8282
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 41.35 on 68 degrees of freedom
## Multiple R-squared:  0.1494, Adjusted R-squared:  0.06187
## F-statistic: 1.707 on 7 and 68 DF,  p-value: 0.1221
```

Appendix Summary: Full Sample vs. Manipulation Check Passers

Interaction Test	b (Full)	p (Full)	N (Full)	b (Pass)	p (Pass)	N (Pass)
RQ1: Gender x Outcome	-16.680	0.298	100	-22.930	0.236	76
RQ3: Gender x Strength	-0.235	0.229	100	-0.261	0.249	76
RQ4: 3-way	-0.103	0.877	100	-0.174	0.828	76